Artifact Appendix: Omniscience for the Masses: New Threats in the Metaverse's Democratized World Creation

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A Artifact Appendix

A.1 Abstract

Our artifact[1] is a single archive of about 90 MB that collects the materials supporting the paper. It contains the video demonstrations of the five omniscience attacks of §4 of the paper, recorded from the attacker and from the victim perspective on Roblox, VRChat, Spatial, Frame, and Horizon Worlds, together with a website that plays each pair side by side with timestamped captions. It further contains the raw and coded data of the platform survey (§3 and Appendix C of the paper), the coding of the 60 papers of the literature survey (§5), the victim-side overhead measurements behind the stealthiness claim of §4.6, and the 3D object we used to implement the one-way material of §4.4. As explained in Appendix B of the paper, we do not release attack scripts or malicious world files. The artifact is released under the CC BY 4.0 license and was awarded the *Artifacts Available* and *Artifacts Evaluated–Functional* badges.

A.2 Description & Requirements

The `README.md` at the root of the archive serves as evaluation guide. It describes every file and indicates which part of the paper each file supports. The archive is organized as follows. The website consists of `index.html`, `script.js`, `styles.css`, and the icons in `media/`; the same website is hosted at https://andrea-mengascini.github.io/worldcreator-artifact/. The `video/` folder holds 34 MP4 files, about 90 MB in total, named `<platform>_<attack>_<attacker|victim>.mp4`, one pair for each platform and attack combination marked as feasible in Table 2. The `csv/` folder holds the platform survey, i.e., the raw results of the Google, Meta Quest Store, Steam Store, VRDB, and SteamDB queries (files starting with `1_`), the 561 merged candidates with their inclusion criteria and popularity scores and the 38 shortlisted applications (files starting with `2_`), and the coding of the world creation tools of each platform behind Tables 1 and 3 (files starting with `3_`). The same folder holds the 60 coded papers of the literature survey (`4_literature_survey.csv`) and, in `benchmark/`, the three runs of the overhead measurements (`run_1.csv` to `run_3.csv`). Finally, `unidirectional_transparent_material.glb` is the one-way material object, and `LICENSE` and `.zenodo.json` hold the license and the Zenodo metadata.

[1] https://doi.org/10.5281/zenodo.22670790

A.2.1 Security, privacy, and ethical concerns. The artifact contains only data, i.e., videos, CSV files, a 3D model, and a web page without server-side components or external requests. We recorded all videos in private instances using our own test accounts; no other users or personal data appear in them. Following the responsible disclosure process described in Appendix B of the paper, we deliberately excluded attack scripts, world files, and any other executable component. Some videos contain speech, and the captions warn the viewer before loud segments.

A.2.2 How to access. The archived version is on Zenodo at https://doi.org/10.5281/zenodo.22670790, a DOI that resolves to the latest release. The maintained version, together with the hosted website, is on GitHub at https://github.com/andrea-mengascini/worldcreator-artifact.

A.2.3 Hardware dependencies. None.

A.2.4 Software dependencies. A modern web browser is sufficient; we recommend Chrome. Python 3 is needed only to serve the website locally. A spreadsheet application helps with the CSV files, and Blender (3.x or later) or a browser-based glTF viewer can open the `.glb` model.

A.2.5 Benchmarks. None.

A.3 Set Up

A.3.1 Installation. No build step is required. After unpacking the archive (or cloning the repository), start a local web server from the folder containing `index.html`, bound to localhost:

```
python3 -m http.server -b localhost 8000
```

The website is then reachable at http://localhost:8000/. Opening `index.html` directly from the file system is not sufficient, as the videos play without sound. The hosted website can be used instead of the local server.

A.3.2 Basic test. Open the website in Chrome, select Roblox, and play the Parabolic Microphone attack. Both videos should start together, pause at each caption, and show a ring around the frame that acts as a volume meter. If a pair does not start, which we observed occasionally in Safari, reload the page or open the two files in `video/` directly. As a second check, `csv/3_result_fro`

`m_the_creation_survey.csv` should contain 25 rows, one per platform of Table 3, and `csv/4_literature_survey.csv` should contain 60.

A.4 Evaluation Workflow

A.4.1 Major claims.

(C1): The five omniscience attacks of §4 can be mounted with standard world creation tools on the platforms marked as feasible in Table 2, and the victim perceives no difference. This claim is supported by (E1) and (E2).

(C2): The attacks impose a negligible overhead on the victim client. On average, frame rate changes by −1.0% and GPU utilization by −1.5%, while CPU utilization rises by about 10% (§4.6). This claim is supported by (E3).

(C3): The platform survey of §3 identifies 25 candidate platforms, which we categorize by editor model and logic paradigm in Tables 1 and 3 and Appendix C. This claim is supported by (E4).

(C4): The literature survey of §5 covers 60 papers, 32 of which propose attacks that fall into seven categories (Appendix D of the extended version). This claim is supported by (E5).

A.4.2 Experiments.

(E1): Attack demonstrations [30 person-minutes]. This experiment supports (C1).
Preparation: Open the website as described in §A.3.1.
Execution: Select each platform in turn and play every attack listed for it. The pairs available on the website correspond to the feasible cells of Table 2, i.e., all five attacks on Roblox and VRChat, Control Room, Astral Projection, and Unidirectional Material on Spatial, and Astral Projection and Unidirectional Material on Frame and Horizon Worlds, for a total of 17 pairs.
Results: In each pair, the attacker video shows the attack taking effect, e.g., the remote audio being captured, the hidden camera feed, the one-way wall, or the replaced voice, while the synchronized victim video shows no visual or auditory cue.

(E2): One-way material [5 person-minutes]. This experiment supports (C1).
Preparation: Open Blender and import `unidirectional_transparent_material.glb` (File, Import, glTF 2.0), then switch the viewport to Material Preview shading.
Execution: Orbit the view around the plane, looking at it first from the front and then from the back.
Results: The plane is visible from the front and disappears from the back, i.e., it behaves as the one-way wall of the Unidirectional Material attack of §4.4.

(E3): Victim-side overhead [10 person-minutes]. This experiment supports (C2).
Preparation: Open the three files in `csv/benchmark/`. Each file reports, per platform and attack, the average frame rate, CPU utilization, and GPU utilization of the victim client without and with the attack, their relative difference, and, on the last line, the average of the run.
Execution: In `run_1.csv`, recompute the average over all attacks and platforms from the per-attack rows. Run 1 is the only run that covers all five platforms, since Spatial discontinued 3D world hosting after it; runs 2 and 3 repeat the measurements on the remaining platforms.
Results: The average of run 1 is −1.0% for frame rate, +9.5% for CPU, and −1.5% for GPU, matching the values reported in §4.6. The largest CPU increase is on Frame, where the world logic runs in the victim's browser.

(E4): Platform survey [15 person-minutes]. This experiment supports (C3).
Preparation: Open `2_App_survey.csv`, `2_selected_app_from_survey.csv`, and `3_result_from_the_creation_survey.csv` in `csv/`.
Execution: Verify that `2_App_survey.csv` lists 561 candidates, merged from the query results in the files starting with `1_`, with the four inclusion criteria, the review and visitor counts, and the z-score, and that `2_selected_app_from_survey.csv` lists the 38 shortlisted applications in ranked order. Then compare `3_result_from_the_creation_survey.csv` with Table 3.
Results: The first 25 shortlisted applications are the platforms of Table 3. Their editor models and logic paradigms, as well as the discarded platforms and the reasons for their exclusion, match Table 3 and the summary in Table 1.

(E5): Literature survey [10 person-minutes]. This experiment supports (C4).
Preparation: Open `csv/4_literature_survey.csv`.
Execution: Count the rows, then filter on the "Attack Group" and "Re-implemented" columns.
Results: The file lists 60 papers, 32 of which carry an attack group within the seven categories of §5. The five attacks re-implemented in §5.1 are marked in the "Re-implemented" column.

A.5 Notes on Reusability

The survey files can be extended with new platforms or papers by adding rows with the same columns. The website has no dependencies, so a new demonstration only requires adding an attacker and a victim video to `video/` and registering them in `index.html`. The one-way material is a single plane with backface culling enabled and can be imported as is into any engine or platform editor that accepts glTF assets.

A.6 Version

Based on the LaTeX template for Artifact Evaluation V20220926.